

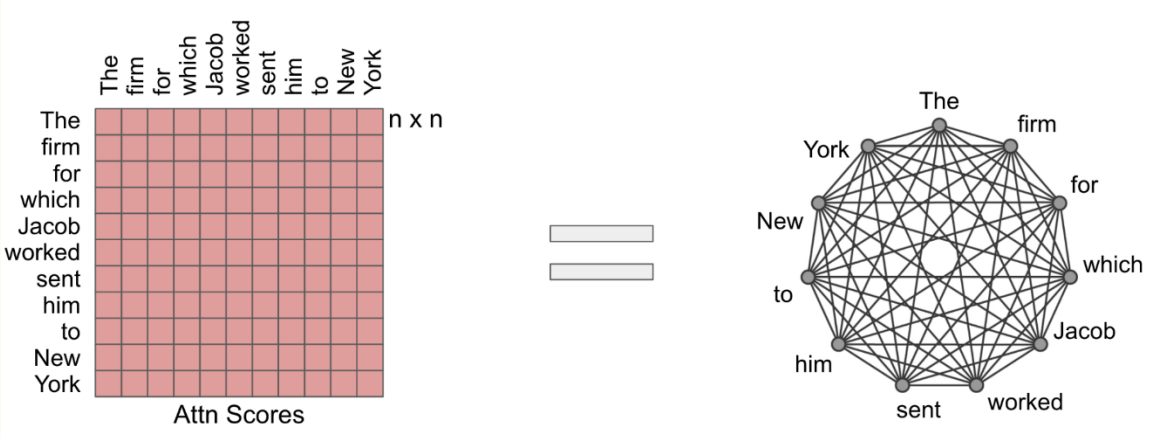

# Introduction to Rag and Langchain

## Demo Task

# Task 0 : Research about BigBird Model



Research the BigBird Transformer model and explain its architecture, motivation, and key innovations. Focus on how BigBird addresses the quadratic complexity of standard self-attention, its sparse attention mechanism, block-sparse patterns, global tokens, and random attention. Compare its computational complexity with BERT and Longformer, discuss suitable applications for long documents, and summarize its main advantages, limitations, and performance characteristics



# Task 1 : your first RAG application

Build a personal Retrieval-Augmented Generation (RAG) system that creates an intelligent knowledge base from a person's professional information. The system should ingest and process sources such as LinkedIn profiles, GitHub repositories, CVs, resumes, portfolios, projects, certifications, and professional experiences. Documents should be chunked, embedded, and stored in a vector database for semantic retrieval.




When the user asks a question about the person, the system should retrieve the most relevant information and provide an accurate, context-aware answer grounded in the stored sources. The system should also cite or identify the source of retrieved information whenever

possible, reducing hallucinations and improving transparency and reliability.